

BOARD OF INTERMEDIATE EDUCATION, KARACHI

Computer Science XI Lectures Paper – I

H.S.C. Part I · Science General & Humanities Groups

Theory 75 + Practical 25 = 200 (with Paper-II)

- Aligned to BIEK Model Paper 2026
- Definitions, comparison tables, diagrams
- 3-mark and detailed answers in board style
- Solved 2026 model paper at the end

Original exam-oriented lecture notes · 2026 edition

How to use these lectures

These notes are original teaching material written for students of the Board of Intermediate Education, Karachi (BIEK). They follow the topics actually asked in Computer Science Paper-I, including the official Model Paper 2026. They are **not** a photocopy of any Sindh Textbook Board book and are **not** an official BIEK paper.

Paper-I is a **theory** paper. Programming in C / Visual Basic is Paper-II (Class XII).

Marks you must remember

Part	Time	Marks	What to do
Section A — MCQs	20 minutes	15	OMR sheet. All 15 compulsory.
Section B — Short	within 2 h 40 min	30	2026 pattern: attempt ALL short parts.
Section C — Detailed	within 2 h 40 min	30	2026 pattern: attempt ALL long questions (each has OR).
Practical	lab exam	25	Assembling, OS, cables, viva, journal.

EXAM TIP

In Model Paper 2026, Section B says **answer all parts** and Section C says **answer all questions**. Older papers allowed choice (attempt 5 of 15, 2 of 3). Follow the instructions printed on *your* question paper on exam day.

How examiners award marks

1. Start every answer with a **one-line definition**.
2. Then give **numbered points** (not a long paragraph).
3. Draw a **neat labelled diagram** wherever the wording is draw / show / classify.
4. Comparison questions need a **table** with at least four differences.
5. Write full forms when an abbreviation is used (LAN, OSI, BIOS, SMTP).

Contents — Class XI / Paper-I

Lecture	Title	Use it for
1	IT and introduction to computers	IT, barcode, types, generations
2	Hardware and I/O devices	Printers, plotter, hardcopy
3	Memory and storage	RAM/ROM, GB, secondary storage
4	CPU, registers, buses, fetch cycle	Section C registers / fetch
5	Software and translators	Compiler vs interpreter
6	Operating system	Section C OS functions
7	Data communication	Channels, modem, signals
8	Networks, topologies, OSI	OSI diagram, LAN/WAN, devices
9	Internet and data security	Virus, HTML, SMTP, BIOS
10	Boolean algebra (past papers)	Gates and DeMorgan
11	MCQ and 3-mark bank	Last-night drill
12	Solved Model Paper 2026	Timed mock

Syllabus map for Paper–I

BIEK Paper–I tests computer fundamentals, architecture, software, operating systems, data communication, networks and security. The 2026 model paper did **not** ask Boolean algebra, but many past papers did — Lecture 10 is a short revision of gates.

Lecture	Topic	2026 model paper links
1	IT and introduction to computers	IT uses/abuses; barcode
2	Hardware and I/O devices	Hardcopy/softcopy; plotter; printers; OMR/OCR
3	Memory and storage	GB = 1024 MB; magnetic disk; secondary storage (C)
4	CPU, registers, buses, fetch cycle	Fetch cycle; registers (C); SIMM; bus
5	Software and language translators	Compiler vs interpreter; GUI
6	Operating system	OS functions (C)
7	Data communication	Components; analog/digital; modulation; channels (C)
8	Networks, topologies, OSI	LAN devices; OSI (C); network types (C)
9	Internet and data security	Virus; HTML; SMTP; BIOS/CRT/SVGA
10	Boolean algebra (past papers)	DeMorgan, gates — keep ready

LECTURE 01

Information Technology and Introduction to Computers

Paper-I · Foundation unit

BOARD DEFINITION — Computer

A computer is an electronic machine that accepts data (input), processes it according to instructions, stores the result and produces output. It works on the IPO cycle: **Input** → **Process** → **Output** (plus storage).

BOARD DEFINITION — Information Technology (IT)

Information Technology is the use of computers, networking, software and telecommunication to collect, process, store, protect and share information.

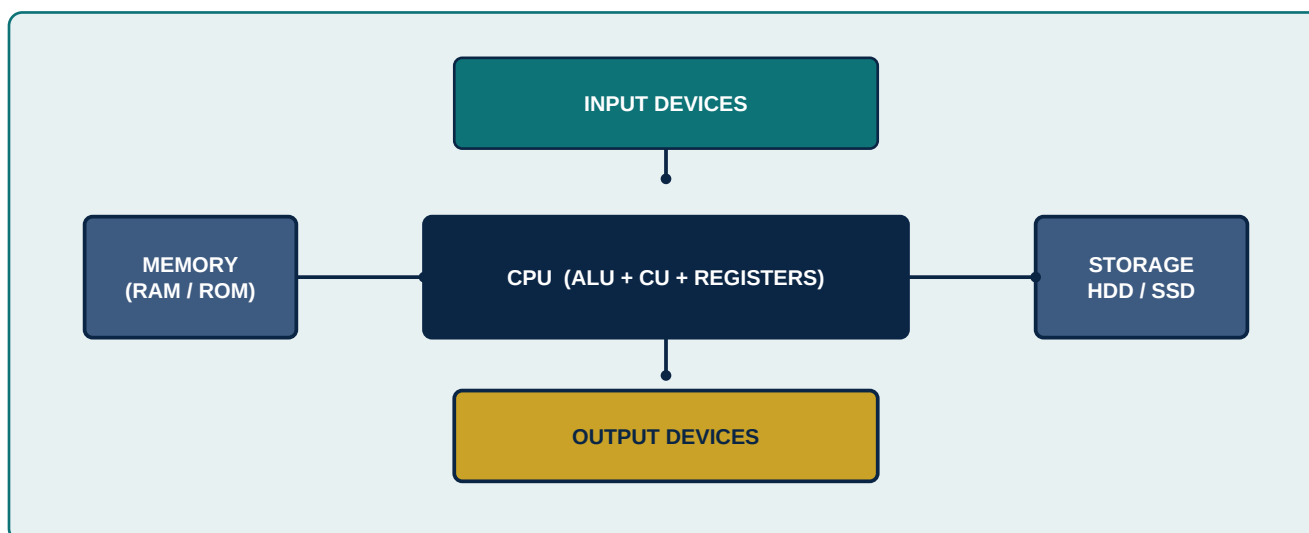


Figure 1.1 Block diagram of a computer system — draw this in long answers.

1.1 Characteristics of a computer

Feature	Meaning for the exam
Speed	Performs millions of instructions per second.
Accuracy	Gives error-free results if the input and program are correct (GIGO).
Diligence	Does not get tired; the 1st and the millionth job are equally accurate.
Storage	Stores huge data on primary and secondary memory.
Versatility	Same machine can do accounts, design, games, research.
Automation	Once programmed, it runs without continuous human help.
No feelings	No emotion, no intelligence of its own (unless AI software is used).

1.2 Advantages of computer education

1. Builds career skills for banking, medicine, engineering, media and e-commerce.

2. Makes office work faster: typing, spreadsheets, presentations, email.
3. Supports online learning, research and digital libraries.
4. Helps in data analysis, graphics, CAD and scientific calculation.
5. Enables communication through internet, video call and social media.

1.3 Uses and abuses of IT

Uses of IT	Abuses of IT
Education: e-learning, LMS, online exams	Cyberbullying and fake news
Health: MRI, patient records, telemedicine	Theft of medical / personal data
Banking: ATM, online transfer, NADRA records	Hacking, phishing, credit-card fraud
Business: inventory, e-commerce, accounts	Software piracy and copyright theft
Government: NADRA, FBR, e-governance	Surveillance without consent; identity theft

1.4 Types of computers

According to size

Type	Point for 3-mark answer
Supercomputer	Fastest; weather, nuclear research, space (e.g. Cray).
Mainframe	Huge storage; banks, airlines, census; many terminals.
Minicomputer	Mid-range; departments and medium organisations.
Microcomputer / PC	Single-user: desktop, laptop, tablet, smartphone.

According to working principle

Type	Data	Example
Analog	Continuous physical quantities	Speedometer, analog thermometer
Digital	Discrete 0 and 1	PC, calculator, digital watch
Hybrid	Mix of analog and digital	Hospital monitoring, petrol pump

1.5 Generations of computers (quick revision)

Gen	Years (approx.)	Technology	Language / remark
1st	1940–56	Vacuum tubes	Machine language; huge, hot, ENIAC
2nd	1956–63	Transistors	Assembly; smaller, more reliable
3rd	1964–71	ICs	High-level languages; minicomputers
4th	1971–present	Microprocessor / VLSI	PCs, GUI, internet
5th	Present +	AI, ULSI, NLP	Robots, expert systems, ML

1.6 Bar code reader

A **bar code** is a pattern of parallel black and white bars that stores a product number (usually UPC / EAN). A **bar code reader** is an optical scanner that reads this pattern with a laser or camera and converts it into a digital code for the computer.

Use in a superstore

1. Cashier scans the item; price and name appear at once from the database.
2. Stock quantity is reduced automatically (inventory control).
3. Billing becomes fast and error-free; no manual price typing.
4. Sales reports and reorder levels can be generated daily.

LIKELY BOARD QUESTION

Q. What is Information Technology? Write any three uses or abuses of IT. (3 marks)

Model answer. IT is the use of computers, software and networks to process and share information. **Uses:** (i) online banking (ii) e-learning (iii) hospital records. **Abuses:** (i) hacking (ii) fake news (iii) software piracy. Write three of either, or a mix, as asked.

LECTURE 02

Computer Hardware and Input / Output Devices

Paper-I · High-frequency unit

BOARD DEFINITION — Hardware

Hardware is the physical, touchable parts of a computer system — keyboard, mouse, monitor, printer, CPU box, cables and chips.

BOARD DEFINITION — Peripheral

A peripheral is any hardware device connected to the CPU to perform input, output or storage, for example keyboard, printer and USB drive.

2.1 Classification of hardware

Class	Job	Examples
Input	Enter data / commands	Keyboard, mouse, scanner, MIC, joystick, OCR, OMR, barcode, webcam, stylus
Output	Show results	Monitor, printer, plotter, speaker, projector, SGD
Processing	Calculate and control	CPU (ALU + CU + registers)
Storage	Hold data	HDD, SSD, USB, SD, CD/DVD, Blu-ray
Communication	Link computers	NIC, modem, router, switch, hub, gateway

2.2 Input devices you must be able to describe

Device	Exam line
Keyboard	Standard QWERTY; alphanumeric, function, numeric and control keys.
Mouse	Pointing device; left/right click, scroll; optical or wireless.
Joystick / joy pad	Gaming and simulation; stick reports X–Y position.
Scanner	Converts paper image into a digital picture (flatbed / handheld).
OCR	Optical Character Recognition — reads printed text into editable type.
OMR	Optical Mark Recognition — reads pencil marks on MCQ sheets.
MICR	Magnetic Ink Character Recognition — bank cheques.
Barcode / QR	Reads product or ticket codes at speed.
Microphone	Voice input; used with speech-recognition software.
Webcam	Live video for calls and recording.
Touch screen	Input + output on the same surface (phones, kiosks).
Digitizer / stylus	Drawing tablets for CAD and graphic design.

2.3 Output devices

Monitors

Type	Working idea	Note
CRT	Electron beam on phosphor screen	Bulky, old, SVGA common in MCQs
LCD	Liquid crystals + backlight	Thin, low power
LED	LCD with LED backlight	Brighter, today's standard
Plasma	Ionised gas cells	Large TVs, high contrast

Softcopy is temporary output on screen or speaker. **Hardcopy** is permanent printed output on paper.

Printers

Impact printers	Non-impact printers
Head hits ribbon / paper	No striking; spray, heat or laser
Noisy	Quiet
Can print carbon copies	No carbon copies
Slower, cheaper (dot matrix, daisy wheel, drum, line)	Faster (inkjet, laser, thermal)
Low print quality (dpi)	High dpi, graphics and colour

Print quality is measured in **DPI (dots per inch)**. Speed is measured in **CPS, LPM or PPM** (characters / lines / pages per minute).

Plotter

A **plotter** is an output device that draws high-quality line graphics using pens. It is used for huge drawings and maps: engineering CAD, architecture, GIS maps, and banners. Types: **drum plotter** and **flatbed plotter**.

Other output

- **Speakers / headphones** — sound output.
- **Multimedia projector** — large-screen display for class and meetings.
- **SGD (Speech Generating Device)** — voice output for users who cannot speak.

LIKELY BOARD QUESTION

Q. What are the differences between Hardcopy and Softcopy? (3 marks)

Model answer. Hardcopy is printed permanent output (paper); softcopy is temporary electronic output (monitor). Hardcopy needs a printer and can be touched; softcopy needs power and can be edited easily. Hardcopy is slower to produce; softcopy is instant. Give 3–4 points in a mini-table.

LECTURE 03

Computer Memory and Storage

Paper-I · MCQ favourite

BOARD DEFINITION — Memory

Memory is the computer's storage area that holds data, instructions and results. It is divided into **primary (main)** memory inside the system unit and **secondary** memory for long-term storage.

3.1 Memory measurement units

Unit	Equals	Board line
Bit	0 or 1	Smallest unit of data
Nibble	4 bits	Half a byte
Byte	8 bits	Stores one character
KB	1024 bytes	Kilobyte
MB	1024 KB	Megabyte
GB	1024 MB	Gigabyte — 2026 MCQ answer
TB	1024 GB	Terabyte
PB	1024 TB	Petabyte

EXAM TIP

BIEK always uses binary multiples: 1 KB = **1024** bytes, not 1000. 2026 MCQ: 1 GB = 1024 MB (option C).

3.2 Primary / main memory

Volatile memory loses contents when power is off. **Non-volatile** memory keeps contents without power.

RAM	ROM
Random Access Memory	Read Only Memory
Volatile	Non-volatile
Read and write	Usually read only (programmed once or rarely)
Holds running programs and data	Holds firmware / BIOS
Faster access; measured in GB today	Smaller; measured in MB
Types: SRAM, DRAM	Types: PROM, EPROM, EEPROM, Flash

RAM types

- **SRAM (Static RAM)** — uses flip-flops; faster; no refresh; used as cache; expensive.
- **DRAM (Dynamic RAM)** — uses capacitors; needs refresh; cheaper; used as main memory.
- **SIMM** — Single Inline Memory Module (32-bit). **DIMM** — Dual Inline Memory Module (64-bit).

ROM types

Type	Can be erased?	How
ROM (mask)	No	Written at factory
PROM	No (once only)	Programmed by user with a PROM burner
EPROM	Yes	Ultraviolet light through a quartz window
EEPROM	Yes	Electrically, byte-wise
Flash	Yes	Electrically, in blocks — USB, BIOS, SSD

3.3 Cache memory

Cache is a small, very fast SRAM between the CPU and RAM. It stores the data and instructions the CPU is most likely to need next. Levels: **L1** (inside core, smallest, fastest), **L2**, **L3** (shared). Cache increases processing speed.

3.4 Secondary storage

Device	Technology	Exam points
Hard disk (HDD)	Magnetic disk	Platters, tracks, sectors, heads; random access; large GB/TB
Floppy disk	Magnetic disk	Obsolete 1.44 MB; still appears in MCQs
Magnetic tape	Magnetic sequential	Backup; cheap; slow sequential access
CD	Optical	700 MB; CD-ROM / CD-R / CD-RW
DVD	Optical	4.7 GB+; movies and software
Blu-ray	Optical (blue laser)	25 GB+ per layer
USB / flash / pen drive	Flash EEPROM	Portable, non-volatile, hot-pluggable
SD / microSD	Flash	Cameras, phones
SSD / external HDD	Flash / magnetic	Fast boot (SSD); backup (external)

Access methods: **Sequential** (tape — start from the beginning) and **Direct / random** (disk, USB — jump to any address).

LIKELY BOARD QUESTION

Q. Explain secondary storage devices with examples. (detailed)

Model answer. Define secondary storage as non-volatile, large, cheaper memory used when the computer is off. Then describe at least four devices with one example each: HDD, optical discs, flash/USB, magnetic tape. Add a comparison: slower than RAM, larger capacity, used for backup and files.

LECTURE 04

Inside the System Unit — CPU, Registers, Buses, Fetch Cycle

Paper-I · Section C favourite

BOARD DEFINITION — System unit

The system unit (CPU cabinet / chassis) is the box that houses the motherboard, power supply, processor, RAM, storage drives, cooling fans and ports.

4.1 Parts of the system unit

- **Power supply (SMPS)** — converts 220 V AC to 12 V / 5 V / 3.3 V DC.
- **Motherboard** — main printed circuit board; everything plugs into it.
- **Microprocessor** — the CPU chip in the socket.
- **RAM slots** — SIMM / DIMM modules.
- **BIOS ROM** — firmware that tests hardware and starts the OS (POST).
- **CMOS battery** — keeps date, time and BIOS settings.
- **Expansion slots** — PCI, PCIe (older papers also mention EISA).
- **Ports** — USB, HDMI, VGA, audio, Ethernet, serial, parallel.
- **Cooling** — heat sink and fan; laptop heat pipes.
- **Drives** — HDD / SSD / optical combo drive.

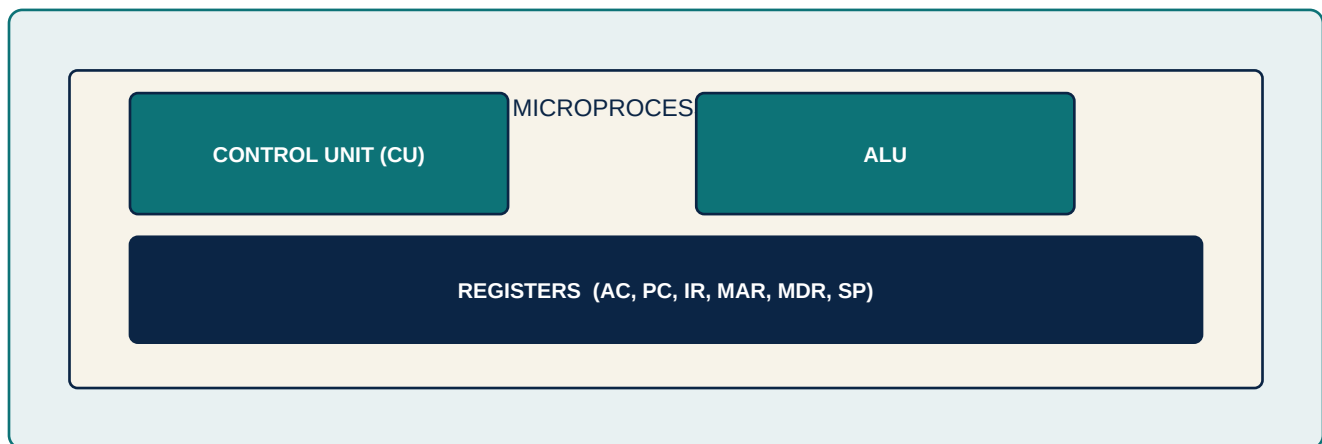


Figure 4.1 Internal view of the microprocessor.

4.2 CPU = ALU + CU + Registers

The **Central Processing Unit** is the brain of the computer. It fetches instructions from memory, decodes them and executes them.

Part	Function
ALU	Arithmetic (+ − × ÷) and logic (AND OR NOT compare) operations.
CU	Controls and coordinates all parts; generates timing and control signals.
Registers	Very fast storage cells inside the CPU for data, addresses and instructions.

Part	Function
Clock	Crystal oscillator; speed in GHz; one cycle = one basic tick.
Cache	On-chip fast memory for hot data.

4.3 Computer registers (draw and explain)

BOARD DEFINITION — Register

A register is a small, high-speed storage location inside the CPU that holds data, an instruction or an address during processing. Registers are faster than cache and RAM.

Register	Full form / role
AC / Accumulator	Holds the result of ALU operations.
PC / Program Counter	Holds the address of the next instruction.
IR / Instruction Register	Holds the instruction currently being executed.
MAR / Memory Address Register	Holds the address of the memory cell to be accessed.
MDR / MBR	Memory Data / Buffer Register — data going to or from memory.
SP / Stack Pointer	Address of the top of the stack.
Flag / Status	Bits that show zero, carry, sign, overflow after ALU work.
Index / General purpose	AX, BX, CX, DX in Intel-style papers — temporary data.

4.4 Fetch cycle (instruction cycle)

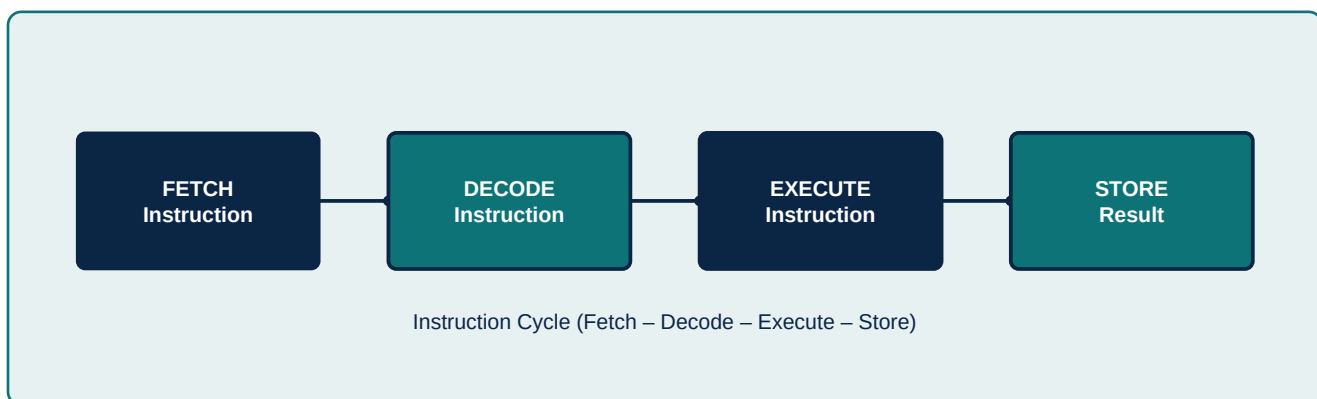


Figure 4.2 Instruction cycle — 2026 short question: 'What is fetch Cycle? Draw its diagram.'

- 1. Fetch:** PC gives address to MAR. Instruction is copied from memory to MDR, then to IR. PC is incremented.
- 2. Decode:** CU reads IR and decides which operation and which operands.
- 3. Execute:** ALU or CU carries out the operation (add, move, jump, I/O).
- 4. Store / Write-back:** Result is written to a register or memory if needed.

4.5 Buses

BOARD DEFINITION — Bus

A bus is a shared electrical pathway that carries data, addresses and control signals between the CPU, memory and I/O devices.

Bus	Carries	Direction
Data bus	Actual data and instructions	Bi-directional
Address bus	Memory / I/O addresses	Uni-directional (CPU → memory)
Control bus	Read, write, interrupt, clock, reset	Mostly uni-directional from CU

Bus **width** (8, 16, 32, 64 bit) decides how many bits travel together. Wider bus = faster transfer.

4.6 Ports

Port	Use
Serial	One bit at a time; old mouse/modem (RS-232).
Parallel	Byte at a time; old printers (Centronics).
USB	Universal Serial Bus — hot plug, keyboards, disks, 2.0/3.x/C.
HDMI	High-Definition Multimedia Interface — digital video + audio.
VGA	Analog video to CRT/old monitors (appears as SVGA in MCQs).

LIKELY BOARD QUESTION

Q. What is Computer Register? Explain its types. (detailed)

Model answer. Define register. Explain at least five: AC, PC, IR, MAR, MDR with one line each. Add: registers are inside CPU, faster than RAM, measured in bits (16/32/64). Draw a small CPU diagram showing registers.

LECTURE 05

Computer Software and Language Translators

Paper-I · Definitions + compiler/interpreter

BOARD DEFINITION — Software

Software is a set of programs, procedures and documentation that tells the hardware what to do. It is intangible — you cannot touch it.

5.1 Types of software

Type	Job	Examples
System software	Controls hardware and provides a platform	OS, device drivers, BIOS, translators
Application software	Solves user tasks	Word, Excel, Chrome, Photoshop, games
Utility software	Housekeeping / maintenance	Antivirus, zip, backup, disk defrag, recovery

Application software

- **General purpose** — used in many fields: word processor, spreadsheet, DBMS, browser, presentation.
- **Special purpose** — written for one job: airline reservation, hospital LIS, NADRA software, ATM software.

5.2 Language translators

BOARD DEFINITION — Language translator

A translator is system software that converts a program written in a programming language into machine language (0s and 1s) that the CPU can execute.

Translator	Converts	How it works
Assembler	Assembly → machine	One-to-one mnemonic translation
Compiler	High-level → machine (object code)	Translates the whole program first, then runs
Interpreter	High-level → machine	Translates and executes line by line

Compiler is more efficient than interpreter — 2026 question

1. Compiler translates the program once; later runs are at full machine speed. Interpreter re-translates every run.
2. Compiled object code can be saved and distributed without source. Interpreted programs need the interpreter present.
3. Compilers perform global optimisation; interpreters generally do not.
4. After compilation, execution does not stop at each line, so run-time is faster.
5. Note: interpreters are easier for beginners (immediate error on the current line). Mention this as a contrast.

5.3 Programming language levels

Level	Example	Point
Machine	01001101	CPU native; difficult for humans
Assembly	MOV AX, 5	Mnemonics; needs assembler
High-level	C, Java, Python, VB	English-like; needs compiler/interpreter
4GL / query	SQL	What to do, not how

5.4 Device drivers and firmware

- **Device driver** — small program that lets the OS talk to a specific hardware device (printer, GPU, NIC).
- **Firmware / BIOS** — software stored in ROM/flash on the board; runs POST and locates the boot device.
- **Anti-malware** — antivirus, anti-Trojan; scans and quarantines malicious programs.

LECTURE 06

Operating System

Paper-I · Almost every year in Section C

BOARD DEFINITION — Operating system

An operating system is system software that manages computer hardware and software resources and provides a platform on which application programs run. It is the interface between the user and the hardware.

Common OS: **Windows, macOS, Linux/UNIX, DOS, Android, iOS.**

6.1 Objectives of an OS

1. Make the computer convenient to use (GUI / commands).
2. Use CPU, memory and devices efficiently.
3. Provide security and access control.
4. Hide hardware details from the user (abstraction).
5. Allow many programs / users to share the machine fairly.

6.2 Functions of an OS (write these in the long question)

Function	What the OS does
Booting	Loads itself from disk into RAM after BIOS POST (cold / warm boot).
Process management	Creates, schedules, suspends and terminates processes; CPU scheduling.
Memory management	Allocates and frees RAM; virtual memory / paging.
File management	Creates folders, keeps FAT/NTFS directories, access rights.
I/O / device management	Talks to devices through drivers; spooling for printer.
Secondary storage management	Disk space, partitions, defragmentation.
Network management	TCP/IP stack, sharing, user login on LAN.
Security / protection	Passwords, permissions, firewall hooks.
Command interpreter / GUI	CMD, PowerShell, bash, or icons and windows.
Error detection	Reports disk fail, no paper, invalid instruction.

6.3 Types / features of OS

Type	Idea	Example
Batch	Jobs collected, run without user sitting there	Old payroll jobs
Multi-tasking	Several programs in memory; CPU switches fast	Windows, Linux
Time-sharing	CPU time slices among many users	UNIX lab servers
Multi-processing	Two or more CPUs / cores share jobs	Modern PCs

Type	Idea	Example
Parallel	Many processors on one task	Scientific clusters
Distributed	Many computers appear as one system	Google data centres
Embedded / real-time	Inside a device; strict time limits	ATM, car ABS, Android TV
Single-user	One person	MS-DOS, home Windows
Multi-user	Many logins	Linux server, mainframe

6.4 Process and its states

BOARD DEFINITION — Process

A process is a program in execution. It includes the code, data, stack and the current register values (PCB).

- **New** — being created.
- **Ready** — waiting for CPU.
- **Running** — using the CPU.
- **Waiting / blocked** — waiting for I/O or an event.
- **Terminated** — finished or killed.

Thread is a lightweight sub-process that shares the same memory space. **Multi-threading** = many threads in one process. **Multi-tasking** = many processes on one OS.

6.5 GUI vs CLI

GUI (Graphical User Interface)	CLI (Command Line Interface)
Icons, windows, menus, pointer	Typed commands
Easy for beginners	Faster for experts / scripts
Windows, macOS, GNOME	DOS, bash, PowerShell

Useful Windows tools that appear in practicals: **Control Panel**, file attributes (R/H/S/A), search, **msconfig** (startup), **dxdiag** (DirectX / hardware report).

LIKELY BOARD QUESTION

Q. What is an Operating System? Explain its functions. (detailed — 2026 Q.3)

Model answer. Write the definition, name 2–3 examples, then explain 8 functions from the table with one sentence each. A small labelled diagram 'User → OS → Hardware' scores extra. This is the safest Section C choice.

LECTURE 07

Data Communication

Paper-I · Heavy in 2026 model paper

BOARD DEFINITION — Data communication

Data communication is the exchange of digital or analog data between two or more devices through a transmission medium and following a set of rules called a protocol.

7.1 Components of data communication (2026 short question)

Component	Role
Sender / source	Device that sends the message (computer, phone).
Receiver / destination	Device that accepts the message.
Message	The information: text, number, picture, audio, video.
Medium / channel	Path: wire, fibre, radio, microwave, satellite.
Protocol	Agreed rules of communication (TCP/IP, HTTP, SMTP).

7.2 Analog vs digital signals

Analog	Digital
Continuous wave; infinite values	Discrete; only 0 and 1
Sine wave	Square wave
Human voice on telephone line	Computer data
Easily affected by noise	More noise-resistant; can be regenerated
Measured in frequency (Hz)	Measured in bit rate (bps)

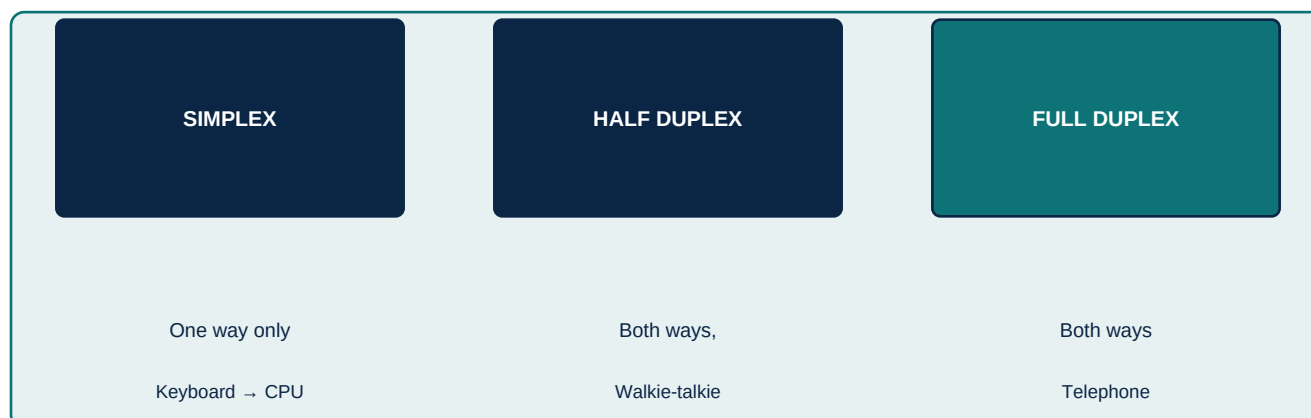
7.3 Communication modes (data flow direction)

Figure 7.1 Simplex, half duplex and full duplex.

Mode	Direction	Example
Simplex	One way only	Keyboard → CPU, radio broadcast, monitor
Half duplex	Both ways, not at the same time	Walkie-talkie, old hub Ethernet
Full duplex	Both ways at the same time	Telephone, modern switched Ethernet

EXAM TIP

2026 MCQ: 'two directions but not simultaneously' = **Half duplex**. Do not confuse with full duplex.

7.4 Serial vs parallel; sync vs async

Serial	Parallel
Bits travel one after another on one wire	A group of bits travel together on many wires
USB, SATA, Ethernet	Old printer port, internal CPU buses
Good for long distance	Good for short distance, high speed internally

Synchronous	Asynchronous
Sender and receiver share a clock	No shared clock; each byte has start/stop bits
Data sent as a continuous block / frames	Data sent character by character
Higher speed; used in networks	Lower speed; old serial terminals, some keyboards
Idle time is filled with idle sync bits	Gap between characters is allowed

7.5 Transmission media / channels (2026 Section C)

Guided (wired)

Medium	Construction	Use / limit
Twisted pair (UTP/STP)	Copper pairs twisted to cut noise; Cat5e/Cat6	LAN, telephone; ~100 m; cheapest
Coaxial	Core, insulation, metal shield, jacket	Cable TV, older LAN; better shield
Fibre optic	Glass/plastic core; light pulses	Backbone, sea cables; fastest, no EMI, costly

Unguided (wireless)

Medium	Point
Radio waves	Omnidirectional; AM/FM, Wi-Fi, Bluetooth, cellular
Microwave	Line of sight; dishes on towers; 1–30 GHz
Infrared	Short range, cannot cross walls; TV remote, old IrDA
Satellite	GEO/MEO/LEO; TV, GPS, remote areas

7.6 Modulation, demodulation and modem

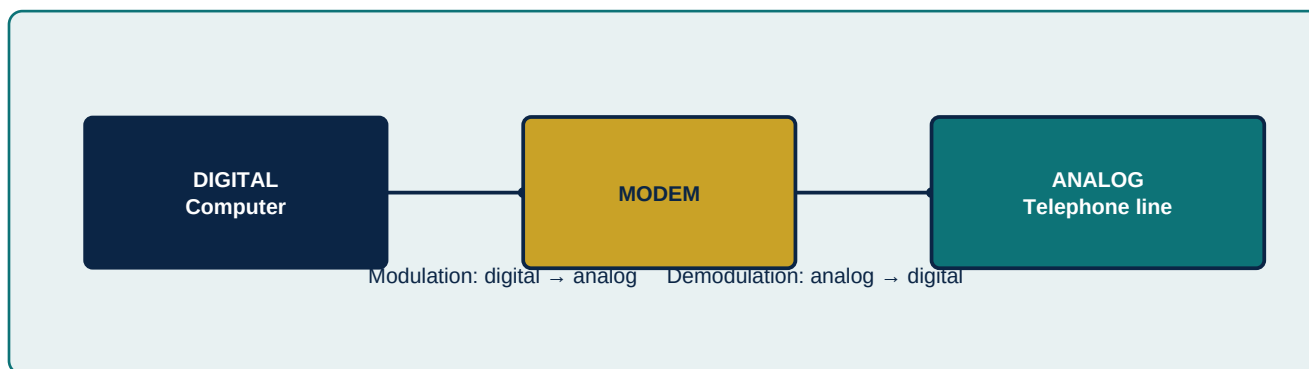


Figure 7.2 Role of a modem.

BOARD DEFINITION — Modulation

Modulation is converting a digital signal into an analog signal so that it can travel on an analog telephone line or radio wave. Methods: AM, FM, PM (amplitude, frequency, phase).

BOARD DEFINITION — Demodulation

Demodulation is converting the received analog signal back into digital form for the computer.

Modem = **Modulator** + **Demodulator**. It sits between a digital computer and an analog line.

EXAM TIP

Standard theory: digital → analog = modulation; analog → digital = demodulation. Some old MCQs wrongly label analog → digital as modulation. On the paper, pick the option that matches **your textbook wording** if two options fight; otherwise use the standard definition. The 2026 MCQ wording 'analog signal to digital signal' is demodulation / ADC — not modulation.

7.7 Broadcast vs point-to-point

Broadcast	Point-to-point
One sender, many receivers on a shared medium	Dedicated link between two nodes
Radio, TV, bus network, Wi-Fi beacon	Telephone call, PPP, most router links
Address may be 'all stations'	Only the two ends see the data

7.8 Coding schemes (short notes)

- **ASCII** — American Standard Code for Information Interchange; 7-bit (128 characters), extended 8-bit.
- **EBCDIC** — Extended Binary Coded Decimal Interchange Code; 8-bit; IBM mainframes.
- **BCD** — Binary Coded Decimal; each decimal digit stored in 4 bits.
- **Unicode / UTF-8** — world languages including Urdu (not always in old papers).

LECTURE 08

Computer Networks, Devices, Topologies and OSI

Paper-I · Highest Section C weight

BOARD DEFINITION — Computer network

A computer network is a collection of two or more computers and devices connected by a medium so that they can share data, hardware and software, following protocols.

8.1 Advantages of networking

1. Resource sharing — printers, files, internet.
2. Communication — email, chat, video.
3. Central backup and security control.
4. Cheaper than giving every user a full set of peripherals.
5. Distributed processing and remote work.

8.2 Types of networks by scale (2026 Section C)

Type	Span	Limitation / note
PAN	A few metres (Bluetooth, USB)	Personal devices only
LAN	Room, building, campus	Privately owned; high speed; limited distance
MAN	A city (cable TV, campus city-wide)	Costly infrastructure; municipal scale
WAN	Country / world (internet)	Lower speed than LAN; uses telecom / satellite; public or leased
Internet	Network of networks	Needs ISP, TCP/IP, global addressing
Intranet	Private TCP/IP network of an organisation	Staff only; may use internet technology internally
Extranet	Intranet opened to partners	Suppliers / customers with login

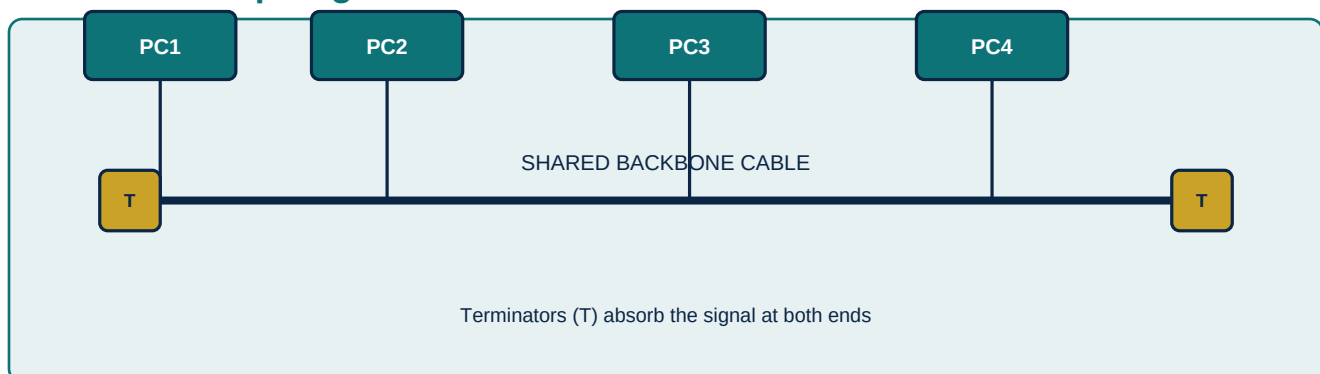
8.3 Network topologies

Figure 8.1 Bus topology — cheapest; one backbone; terminators required. 2026 MCQ.

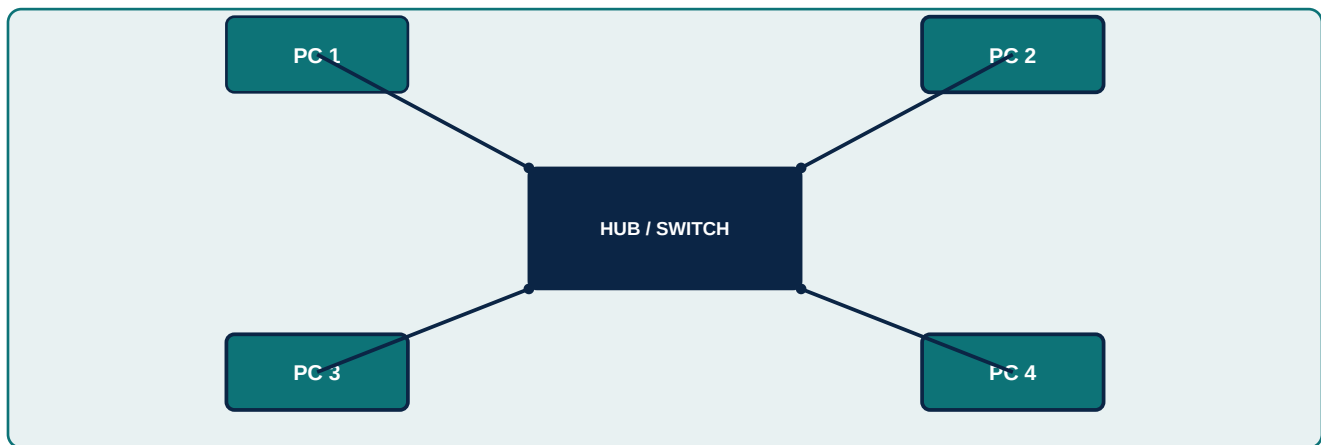


Figure 8.2 Star topology — each node to a hub/switch; easy fault finding.

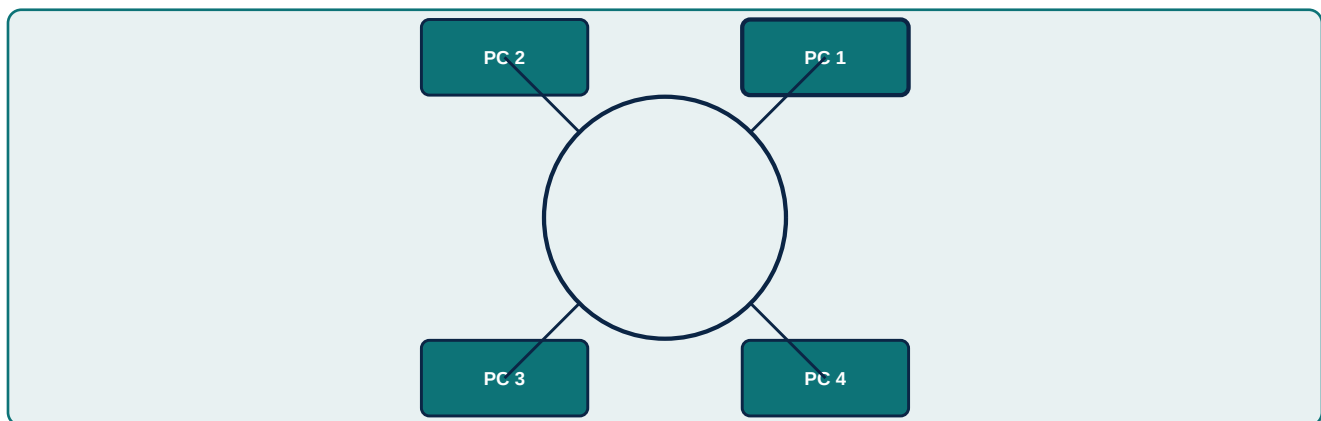


Figure 8.3 Ring topology — token passing; a break can stop the ring (unless dual ring).

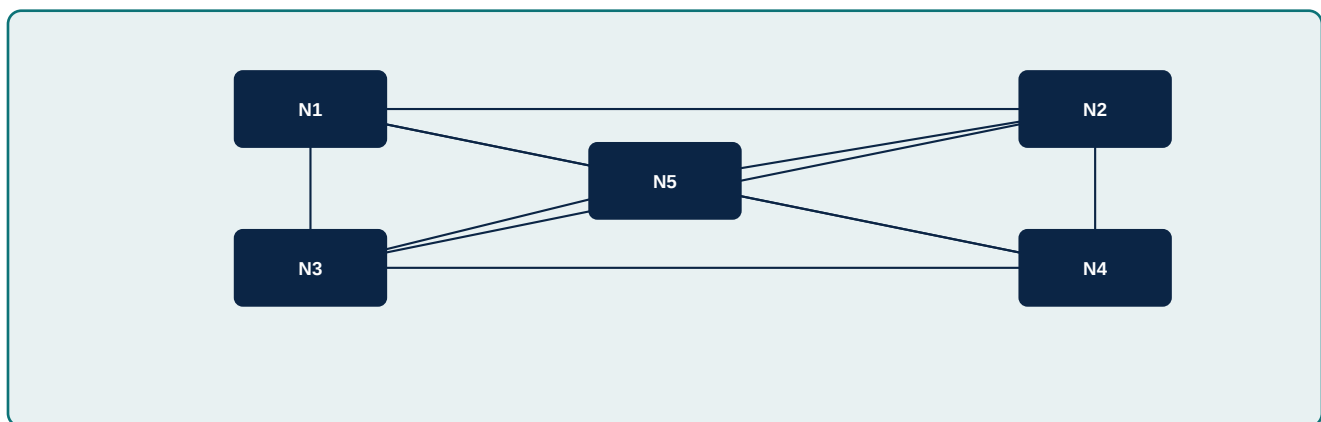


Figure 8.4 Mesh — every node linked to many others; most reliable, most cables.

Topology	Plus	Minus
Bus	Cheap, short cable	Collision; backbone break kills all; hard to find fault
Star	Easy add/remove; one cable fault affects one PC	Central hub/switch failure kills all; more cable
Ring	Fair access (token)	A cut can stop network; harder to reconfigure
Mesh	No single point of failure	Expensive; complex
Tree / hybrid	Fits buildings with many floors	Depends on the mix

8.4 Network devices (LAN components)

Device	Layer (OSI)	Job in one line
NIC	1–2	Card that gives a computer a MAC address and connects to the medium
Repeater	1	Regenerates weak analog/digital signals over long cable
Hub	1	Broadcasts bits to all ports (old, collisions)
Switch	2	Forwards frames using MAC table — today's LAN centre
Bridge	2	Connects two similar LANs / segments (2026 MCQ)
Router	3	Connects different networks using IP; chooses path
Gateway	4–7	Connects two different networks / protocols (2026 short Q)
Modem	1	Digital ↔ analog for telephone / cable line
Access point	1–2	Wi-Fi radio that joins wireless clients to a LAN

EXAM TIP

Memorise this pair: **Bridge = two similar networks**. **Gateway = two different networks / protocols**. Router connects different IP networks. Switch is inside one LAN.

8.5 OSI reference model (2026 Section C — draw and explain)

BOARD DEFINITION — OSI

OSI (Open Systems Interconnection) is a seven-layer reference model published by ISO that describes how data travels from one application to another across a network.

7 Application	HTTP, SMTP, FTP, DNS
6 Presentation	Encryption, compression
5 Session	Start / manage / end session
4 Transport	TCP, UDP — ports
3 Network	IP, routing
2 Data Link	MAC, frames, NIC, switch
1 Physical	Cables, hubs, bits

Figure 8.5 ISO-OSI seven layers. Always draw this stack in the long answer.

Memory sentence: **Please Do Not Throw Sausage Pizza Away** (Physical → Application) or the reverse **All People Seem To Need Data Processing**.

What each layer does (write 2 lines each in Section C)

1. **Physical:** bits on wire/radio; voltage, fibre, connectors, hubs, repeaters.
2. **Data link:** frames, MAC addresses, error detection, switches, NIC.
3. **Network:** logical IP addressing and routing; routers; best path.
4. **Transport:** end-to-end reliability (TCP) or fast datagrams (UDP); port numbers.
5. **Session:** start, manage and close a dialogue between applications.
6. **Presentation:** format, encryption, compression (SSL/TLS often discussed here).
7. **Application:** services the user sees — HTTP, SMTP, FTP, DNS, Telnet.

8.6 TCP/IP model (short)

TCP/IP layer	Close OSI layers	Examples
Application	5, 6, 7	HTTP, SMTP, FTP, DNS
Transport	4	TCP, UDP
Internet	3	IP, ICMP, ARP
Network access / link	1, 2	Ethernet, Wi-Fi

TCP/IP is the protocol suite of the internet. **IP** delivers packets; **TCP** makes the stream reliable.

8.7 Standards bodies

- **ISO** — International Organization for Standardization (OSI model).
- **IEEE** — Institute of Electrical and Electronics Engineers (802.3 Ethernet, 802.11 Wi-Fi).
- **ITU** — International Telecommunication Union (phone and radio standards).
- **IANA / ICANN** — IP addresses and domain names (extra).

LECTURE 09

Internet, Email and Data Security

Paper-I · Virus, HTML, SMTP, BIOS

BOARD DEFINITION — Internet

The internet is a worldwide public network of networks that uses the TCP/IP protocol suite to share information and services.

BOARD DEFINITION — Intranet

An intranet is a private network that uses internet technologies (browsers, HTTP, email) but is available only inside an organisation.

9.1 Important internet terms

Term	Full form / meaning
WWW	World Wide Web — linked hypertext documents viewed in a browser
URL	Uniform Resource Locator — address of a web resource
HTML	HyperText Markup Language — language of web pages (2026 MCQ)
HTTP / HTTPS	HyperText Transfer Protocol (Secure)
ISP	Internet Service Provider
DNS	Domain Name System — name to IP
FTP	File Transfer Protocol
SMTP	Simple Mail Transfer Protocol — sending email (2026 MCQ)
POP3 / IMAP	Receiving email from a mailbox
IP address	Logical 32-bit (IPv4) or 128-bit (IPv6) host identity
MAC address	48-bit burned-in NIC identity

9.2 Email

Electronic mail sends messages to a mailbox over a data network. A message has From, To, Subject, body and optional attachment. Sending uses **SMTP**; receiving uses **POP3 or IMAP**.

9.3 Computer virus and other malware

BOARD DEFINITION — Computer virus

A computer virus is a malicious program that attaches itself to other programs or documents, replicates, and can damage files, slow the system or steal data. It is **software**, not hardware (2026 MCQ).

Malware	Behaviour
Virus	Needs a host file; spreads when the host is copied / run
Worm	Spreads by itself over a network
Trojan horse	Looks useful, hides a harmful payload; does not self-replicate
Spyware / keylogger	Steals passwords and activity
Ransomware	Encrypts files and demands money
Time bomb / logic bomb	Triggers on a date or condition

Antivirus examples (write any five)

Norton, McAfee, Kaspersky, Avast, AVG, Bitdefender, Windows Defender, Avira, ESET NOD32, Quick Heal. Use keeps definitions **updated**.

9.4 Protection methods

1. Install and update antivirus / anti-malware.
2. Use a **firewall** — hardware or software that filters packets between a trusted LAN and the internet.
3. Strong passwords and two-factor authentication.
4. Do not open unknown attachments; beware phishing.
5. Keep OS and browsers patched; take backups.
6. A **proxy** server can hide internal IPs and filter web content.

9.5 Full forms drill (2026 Q.2 x)

Abbrev.	Full form
OCR	Optical Character Recognition
OMR	Optical Mark Recognition
MICR	Magnetic Ink Character Recognition
GUI	Graphical User Interface
CLI	Command Line Interface
CRT	Cathode Ray Tube
LCD / LED	Liquid Crystal Display / Light Emitting Diode
SVGA	Super Video Graphics Array
BIOS	Basic Input Output System
POST	Power On Self Test
USB	Universal Serial Bus
HDMI	High-Definition Multimedia Interface
SIMM	Single Inline Memory Module

LECTURE 10

Boolean Algebra and Logic Gates (past-paper revision)

Paper-I · Not in 2026 model; common earlier

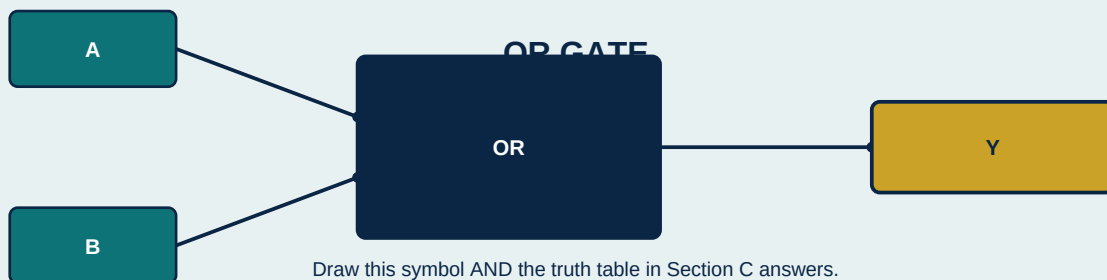
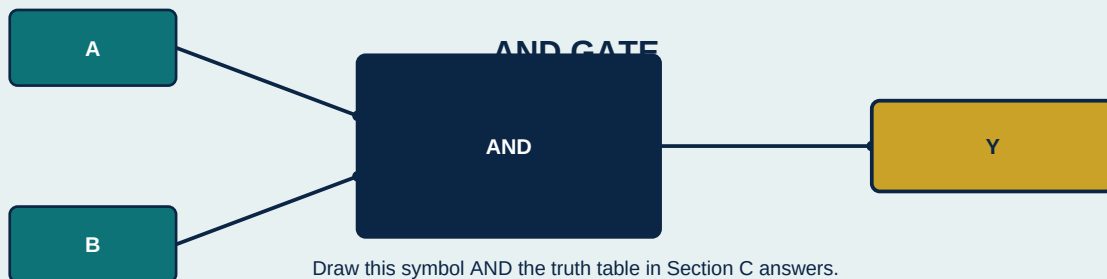
Model Paper 2026 did not include Boolean algebra. Many older BIEK papers did. Revise this lecture if your college still teaches Chapter 4 of the old book.

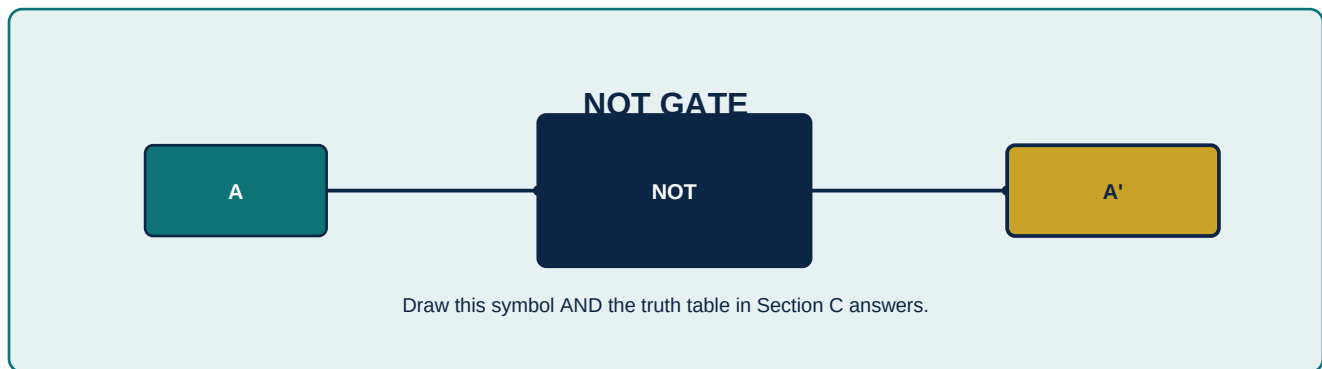
BOARD DEFINITION — Boolean algebra

Boolean algebra is the algebra of logic, developed by George Boole, in which variables take only two values: TRUE/1 and FALSE/0. It is the basis of digital circuits.

10.1 Basic operations

Operation	Symbol	Meaning	Gate
AND (product)	$A \cdot B$ or AB	1 only if all inputs are 1	AND
OR (sum)	$A + B$	1 if any input is 1	OR
NOT (complement)	A' or $\bar{A}$	Inverts $0 \leftrightarrow 1$	NOT
NAND	$(AB)'$	AND then NOT	NAND (universal)
NOR	$(A+B)'$	OR then NOT	NOR (universal)
XOR	$A \oplus B$	1 if inputs differ	XOR
XNOR	$(A \oplus B)'$	1 if inputs are same	XNOR





A	B	AND	OR	NAND	NOR	XOR
0	0	0	0	1	1	0
0	1	0	1	1	0	1
1	0	0	1	1	0	1
1	1	1	1	0	0	0

10.3 Important laws

Law	OR form	AND form
Identity	$A + 0 = A$	$A \cdot 1 = A$
Null	$A + 1 = 1$	$A \cdot 0 = 0$
Idempotent	$A + A = A$	$A \cdot A = A$
Complement	$A + A' = 1$	$A \cdot A' = 0$
Commutative	$A + B = B + A$	$AB = BA$
Associative	$(A+B)+C = A+(B+C)$	$(AB)C = A(BC)$
Distributive	$A+BC = (A+B)(A+C)$	$A(B+C) = AB+AC$
Absorption	$A + AB = A$	$A(A+B) = A$
Double complement	$(A')' = A$	—

DeMorgan's theorems (very common)

1. $(A + B)' = A' \cdot B'$ — NOT of OR is AND of NOTs.
2. $(A \cdot B)' = A' + B'$ — NOT of AND is OR of NOTs.

Always show **both truth tables** and a logic diagram when asked.

LECTURE 11

MCQ drill and 3-mark bank

Paper-I · Practice

11.1 One-line MCQ facts

1. 1 GB = 1024 MB. 1 Byte = 8 bits. 1 Nibble = 4 bits.
2. Floppy and HDD are magnetic disks (not optical).
3. SIMM = Single Inline Memory Module.
4. OSI has seven layers. Cheapest topology is bus.
5. Virus is software. OMR is a scanning input device.
6. Bus is the electrical path among devices.
7. Bridge connects two similar networks; gateway connects different networks.
8. Coaxial and fibre are communication media.
9. HTML produces web pages. SMTP is the common email-sending protocol.
10. Analog → digital for a modem is demodulation. Half duplex = both ways, not together.
11. ALU = Arithmetic and Logic Unit. BIOS = Basic Input Output System.
12. Compiler translates the whole program; interpreter line by line.
13. Laser and inkjet are non-impact; dot matrix is impact.
14. Plotter prints huge drawings and maps.
15. TCP/IP is the internet protocol suite. Fast Ethernet is 100 Base-T.

11.2 Short questions — write 5–7 lines each**Q. Define volatile and non-volatile memory.**

Volatile loses data without power (RAM). Non-volatile keeps data (ROM, HDD, flash).

Q. What is a protocol?

A protocol is a set of rules for communication (format, timing, error handling). Example: TCP/IP, HTTP, SMTP.

Q. Differentiate compiler and interpreter.

Compiler: whole program, object file, faster run. Interpreter: line by line, slower run, easier debug.

Q. What is a gateway?

A gateway connects two networks that use different protocols; it can convert packet formats. It works at higher OSI layers.

Q. Define LAN and WAN.

LAN covers a building, high speed, private. WAN covers cities/countries, uses telecom, the internet is the largest WAN.

Q. What is BIOS?

BIOS is firmware in ROM that runs POST, initialises hardware and loads the boot loader / OS.

Q. Impact vs non-impact printer (four differences).

Use the table in Lecture 2. Always write four rows.

Q. What is modulation?

Changing digital data into analog form for analog media. Modem performs modulation and demodulation.

Q. Name five antiviruses.

Norton, Kaspersky, Avast, McAfee, Windows Defender.

Q. What is cache memory?

Small fast SRAM between CPU and RAM that stores likely-next instructions/data to reduce wait.

LECTURE 12

Solved BIEK Model Paper 2026 — Computer Science Paper-I

Paper-I · Use as a timed mock

The wording below follows the official Model Paper 2026 (Science General Group). Answers are original model answers, not copied from any guidebook.

Section A — MCQs (15 × 1 = 15)

#	Answer	Why
1	C) 1024 MB	1 GB = 1024 MB
2	C) Magnetic Disk	Floppy and HDD
3	C) Single Inline Memory Module	SIMM
4	D) seven layers	OSI
5	B) Bus	Cheapest topology
6	B) Software	Virus is a program
7	D) OMR	Scanning input
8	A) Bus	Electrical path
9	B) Bridge	Two similar networks
10	B) Communication Media	Coax and fibre
11	B) HTML	Web pages
12	C) SMTP	Sending email
13	C) Demodulation	Analog → digital at the modem (see Lecture 7 tip)
14	B) Protocol	Rules on the internet
15	B) Half duplex	Both ways, not simultaneous

Section B — Short answers (all parts)

Q. 2(i) What is Information Technology? Write any three uses or abuses of IT. OR How can a Bar Code Reader be useful in a superstore?

IT: use of computers and networks to process information. Uses: e-learning, e-banking, telemedicine. OR Barcode: instant price, automatic stock update, fast error-free billing, sales reports.

Q. 2(ii) Hardcopy vs Softcopy OR Synchronous vs Asynchronous transmission

Hardcopy = printed paper; softcopy = screen. OR Sync shares a clock and sends blocks; async uses start/stop bits per character.

Q. 2(iii) Define computer virus with examples. Name any five antiviruses. OR Devices to print huge drawings or maps.

Virus = malicious self-replicating software (ILOVEYOU, Melissa). Antiviruses: Norton, Kaspersky, Avast, McAfee, Defender. OR Plotters (drum and flatbed); large-format inkjet for maps.

Q. 2(iv) What is fetch Cycle? Draw its diagram. OR Name the components of data communication.

Fetch-decode-execute-(store) as in Figure 4.2. OR Sender, receiver, message, medium, protocol.

Q. 2(v) Modulation and Demodulation OR How is a Compiler more efficient than Interpreter?

Modulation digital → analog; demodulation analog → digital; modem does both. OR Compiler translates once to object code so execution is faster; interpreter re-translates every line every run.

Q. 2(vi) Four differences: Impact vs Non-Impact printers.

Strike vs no strike; noisy vs quiet; carbon copies vs none; slower/cheaper vs faster/high dpi. Examples: dot matrix vs laser.

Q. 2(vii) LAN component that allows communication between two different networks.

Gateway (also acceptable in some marking: router if they want IP networks). Best answer: Gateway.

Q. 2(viii) Analog signals vs Digital signals.

Continuous vs discrete; sine vs square; noise-sensitive vs regenerable; voice vs computer data.

Q. 2(ix) Broadcast vs Point-to-Point.

One-to-many shared medium vs dedicated two-end link. Radio vs telephone call.

Q. 2(x) Full form of any three: OCR, GUI, SVGA, CRT, BIOS.

OCR Optical Character Recognition; GUI Graphical User Interface; SVGA Super Video Graphics Array; CRT Cathode Ray Tube; BIOS Basic Input Output System.

Section C — Detailed answers**Q. 3. Operating System and its functions OR Computer Register and its types**

See Lectures 6 and 4. Write definition + 8 functions, or definition + AC, PC, IR, MAR, MDR, flags + diagram.

Q. 4. Data communication channels with examples OR Secondary storage devices with examples

See Lectures 7 and 3. Guided (twisted pair, coax, fibre) and unguided (radio, microwave, satellite, infrared) with one use each. OR HDD, optical, flash, tape with properties.

Q. 5. Draw and explain OSI OR Computer Network classified by scale and limitation

Draw seven boxes, two lines per layer. OR PAN, LAN, MAN, WAN with distance, ownership and speed/cost limits.

Practical journal — Paper-I related

1. Identify motherboard parts: RAM, CPU fan, SATA, 24-pin power, CMOS battery, PCI slots.
2. Show BIOS boot order and POST.
3. Windows: create / copy / rename / delete folders; set Hidden and Read-only attributes.
4. Control Panel: date/time, mouse, uninstall a program.
5. Make a straight-through UTP cable (T568B both ends) and a cross-over cable; test with a LAN tester.
6. Set a static IP, subnet mask and gateway; ping another PC.
7. Share a folder on LAN.

End of Class XI / Paper-I lectures. Continue with the Class XII book for Programming in C (Option I), Visual Basic (Option II) and DBMS / MS-Access, which BIEK examines in Paper-II.